

BSPLINE Inference Comparison

Polishing task -- 0° stain orientation

2026-08-19

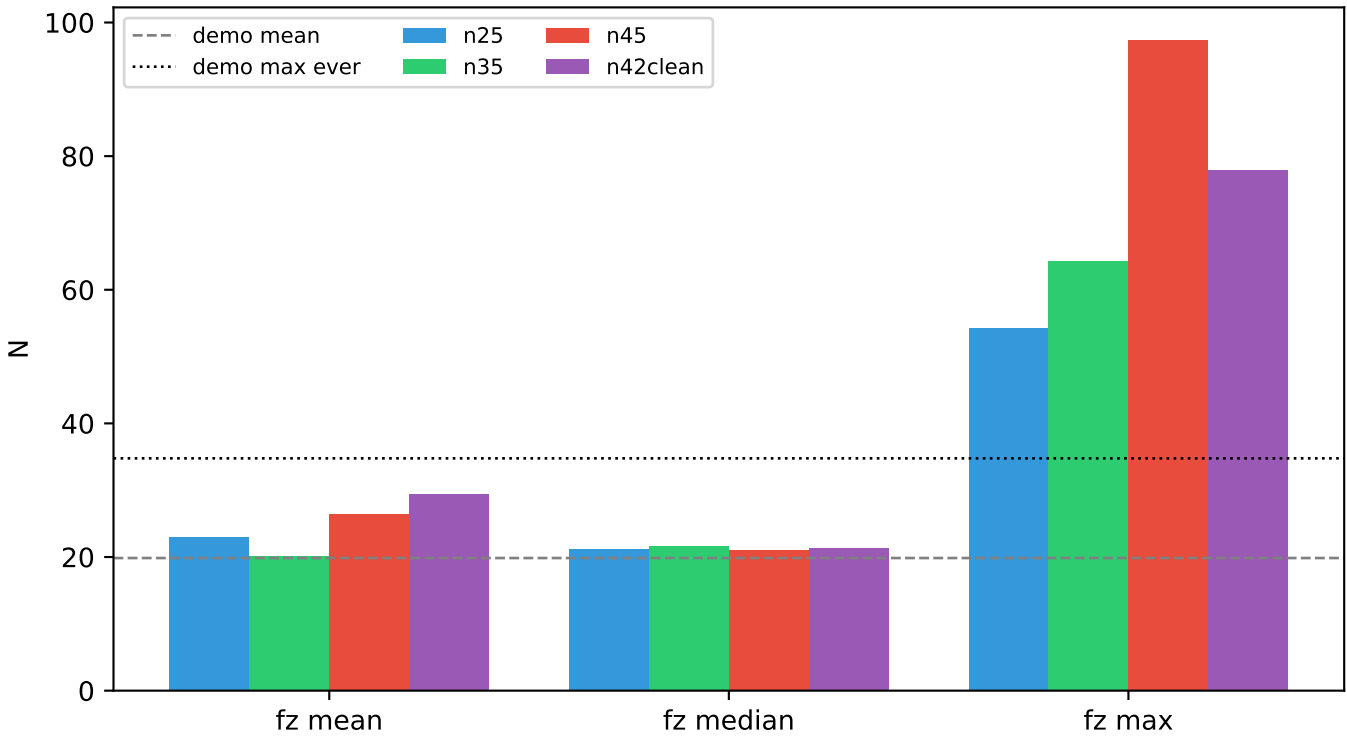
Checkpoints compared (all BSPLINE, FACTR2 phase-resample enabled, tcp_roi_area_fraction=0.25, trained on stain-match-ranked subsets of the corrected 20260817_2046 / 20260817_2128 datasets):

n25	25 episodes (top stain-match rank)	best_epoch=12	val_loss=0.0423
n35	35 episodes	best_epoch=14	val_loss=0.0208
n45	45 episodes (incl. 3 outliers)	best_epoch=103	val_loss=0.0217
n42clean	42 episodes (outliers excluded)	best_epoch=34	val_loss=0.0303

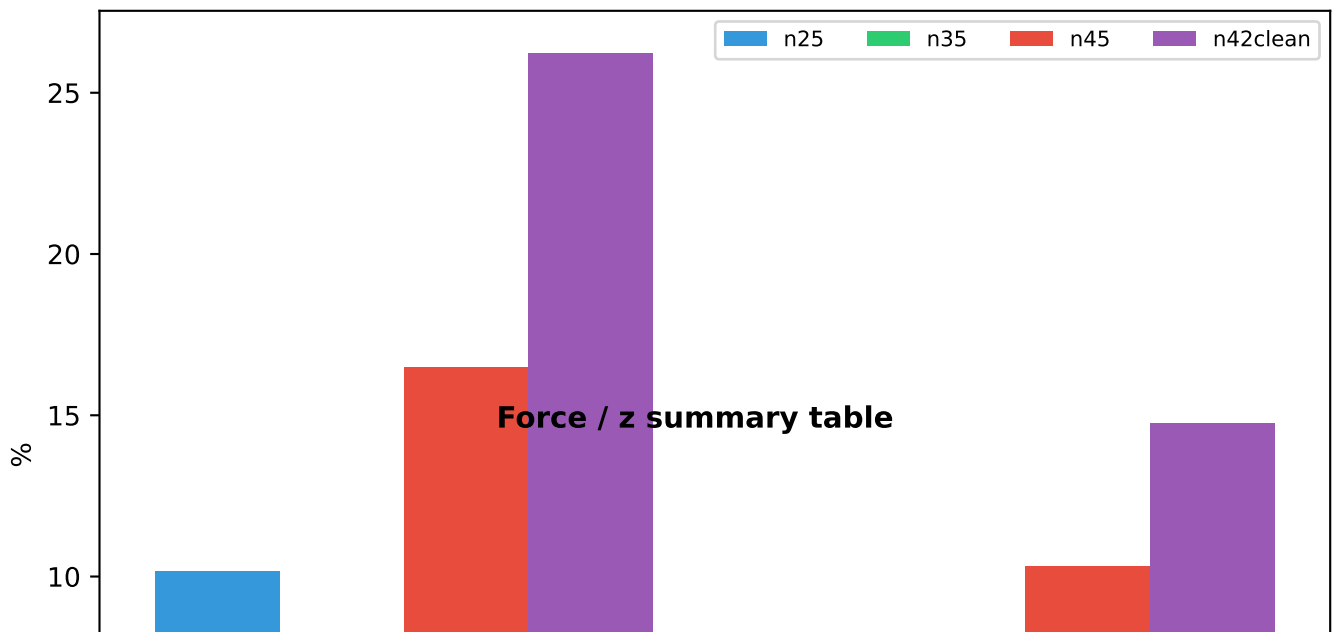
Demo (human-taught) reference, contact-phase, 29-episode corrected set:

fz mean=19.9N median=21.9N max(ever)=34.8N
stain-box contact-time fraction: mean=0.54

Contact-phase force vs. demo reference



Excessive-force fraction (demo: 0% for both)

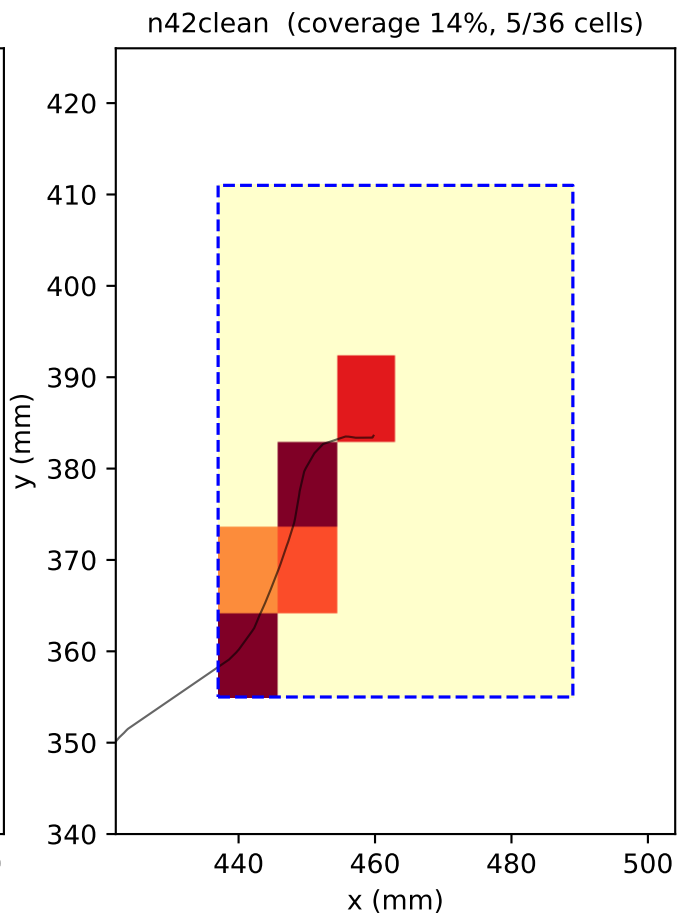
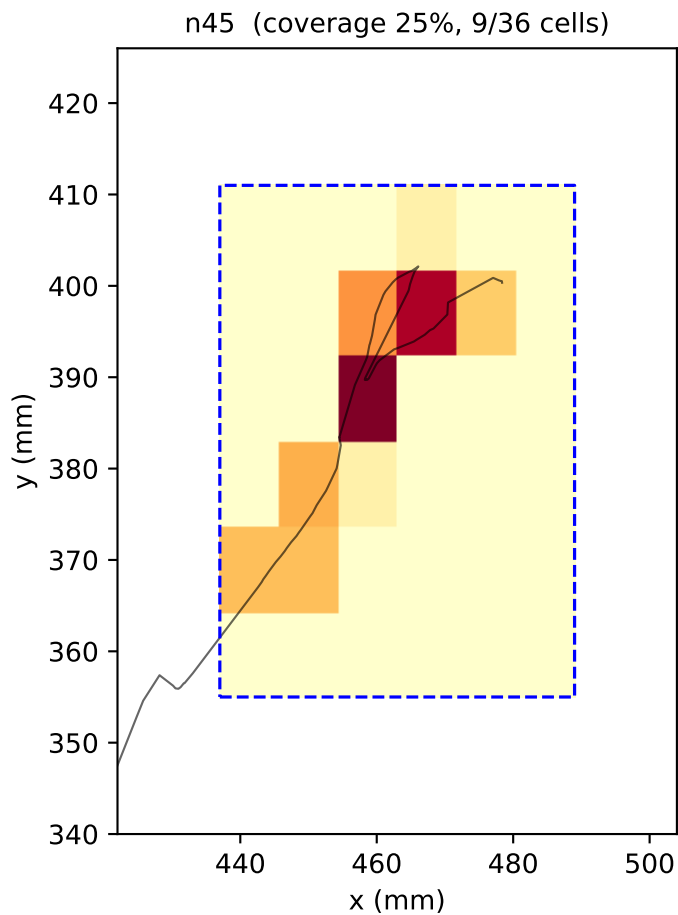
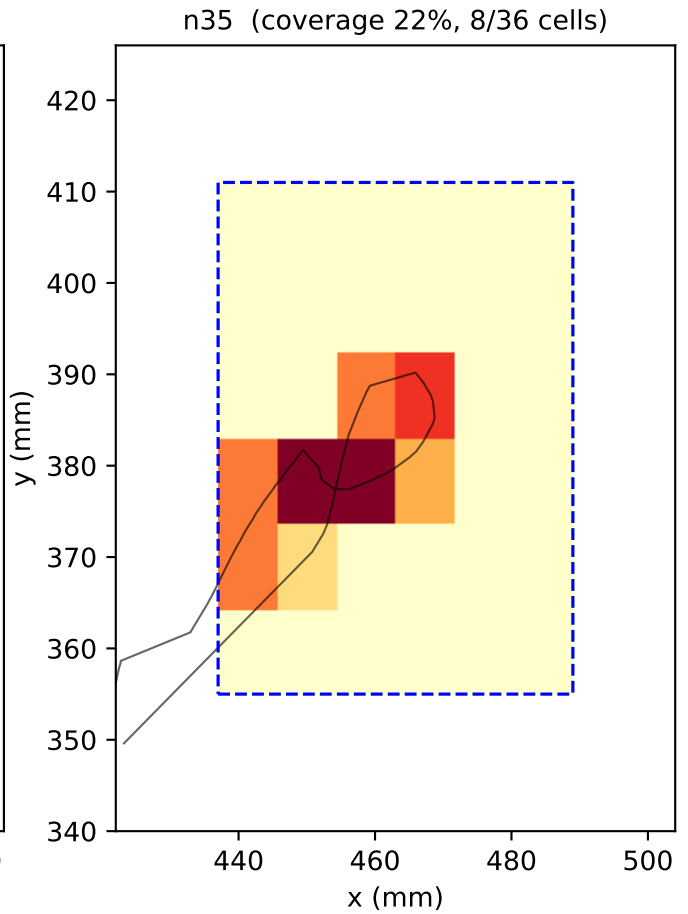
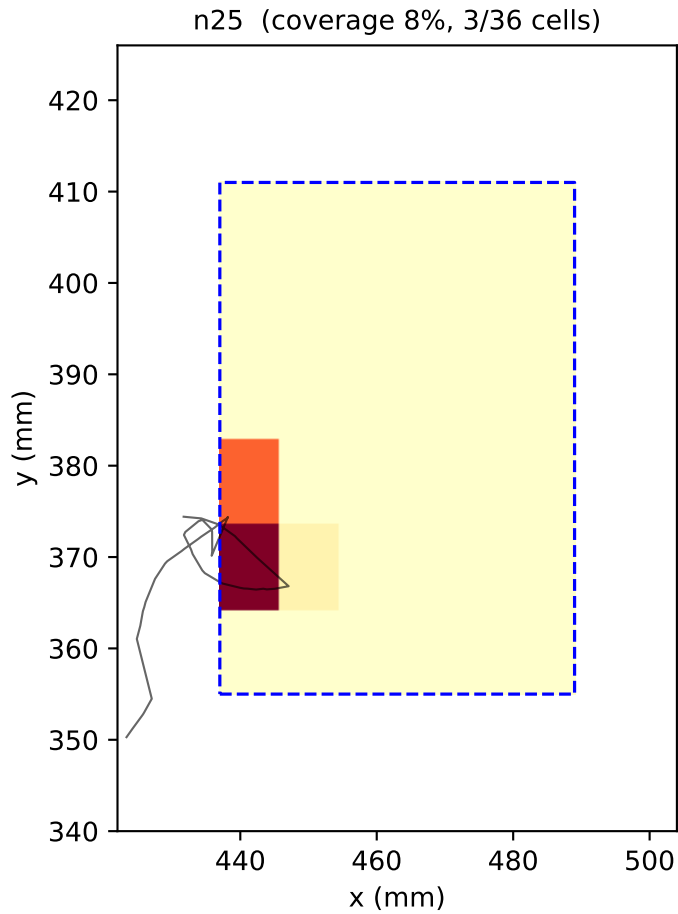


run	fz mean	fz med	fz max	>40N	>60N	z range (mm)
n25	22.9	21.1	54.2	10%	0%	174.7-177.3
n35	20.1	21.6	64.3	2%	1%	174.8-209.4
n45	26.4	21.0	97.4	16%	10%	173.3-177.1
n42clean	29.4	21.3	77.8	26%	15%	173.3-176.2

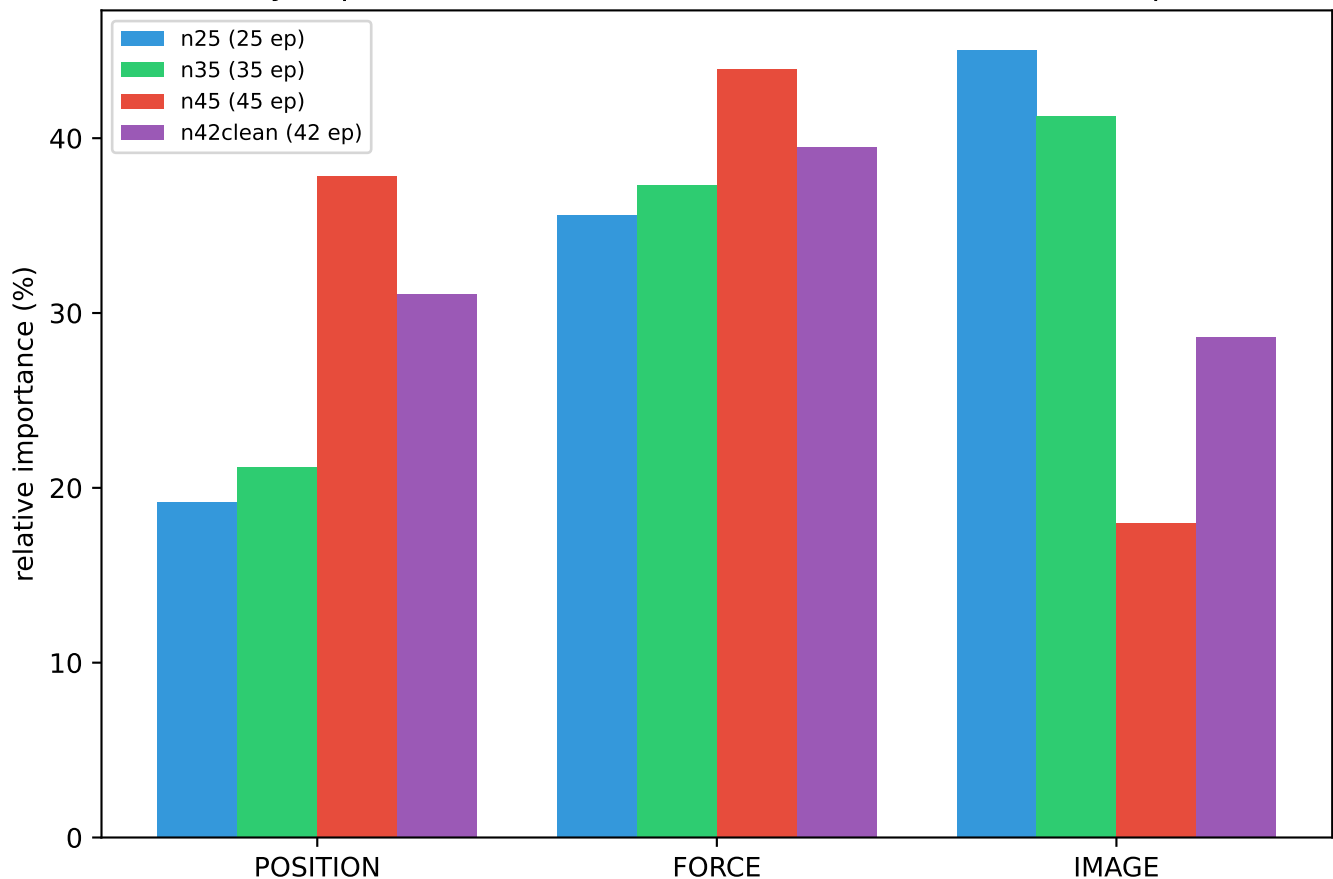
fz > 40N fraction

fz > 60N fraction

Stain-box contact coverage (6x6 grid over (437,355)-(489,411))



Modality-importance ablation (relative share, sums to ~100% per run)



Interpretation notes:

- This is a counterfactual sensitivity diagnostic, NOT proof that force is imitated more accurately. It measures how much the predicted action changes when a modality is ablated -- i.e. how reactive the policy is to that input, not whether its response magnitude is well-calibrated.
- FORCE-importance rises monotonically with dataset size (35.6% -> 37.3% -> 39.5% -> 44.0% for n25 -> n35 -> n42clean -> n45), and this tracks closely with the excessive-force fractions on the previous page -- higher force-sensitivity plausibly amplifies force-loop noise into larger realized contact forces (a control-theoretic "higher gain" effect), rather than reproducing the demo's force profile more faithfully.
- Because this is a normalized 3-way share, a rising FORCE% can also simply reflect IMAGE becoming a less consistent/discriminative signal as the training set grows and diversifies, rather than force becoming more informative in an absolute sense. Distinguishing these needs the raw (un-normalized) ablation scores, not just the rendered bar share.

Trajectory-quality summary

run	net/path	#contact segments	in-box fraction	x reach (mm)	y reach (mm)	grid coverage
n25	0.170	4	34%	423-447	350-374	8%
n35	0.042	7	62%	419-469	339-390	22%
n45	0.083	3	85%	422-478	347-402	25%
n42clean	0.313	2	51%	417-460	343-384	14%

Key findings (2026-08-19, 0° stain):

1. Only n45 fully cleared the stain in this test. It reached furthest into the stain box (x 422-478, y 348-402) with a diagonal sweep covering 25% of the grid -- more than double n25's 8%. The other three checkpoints stayed clustered in the upper-left portion of the box and never physically reached large parts of it.
2. "More training data" alone does not explain this: n42clean (42 ep) covers LESS of the grid (14%) than n35 (35 ep, 22%) despite having more episodes. The one metric that *is* monotonic with dataset size is force-importance (see previous page) -- n45's outcome looks tied specifically to the 3 outlier episodes it uniquely includes, not to raw data volume.
3. n45's better coverage came with a real cost: highest force spikes after n42clean (max 97.4N vs. demo's ever-max of 34.8N, fz>60N in 10% of contact samples). n35 stayed closest to the demo's force profile (fz>40N only 2%) but swept the smallest useful area and broke contact 7 separate times.
4. Caveat: each checkpoint was tested with a SINGLE inference run. Run-to-run variance (start pose, minor calibration drift, timing) is not yet characterized -- these rankings should be confirmed with 2-3 repeats per checkpoint before treating them as settled.

Recommended next steps

1. Repeat inference 2-3x per checkpoint (same 0° stain) to separate real model differences from run-to-run noise before picking a 'winner'.
2. If n45 is the preferred direction, investigate whether its force spikes (max 97.4N) are acceptable for the actual polishing pad/workpiece, or whether the outlier-episode influence needs to be tempered (e.g. down-weighting those 3 episodes rather than a hard include/exclude choice).
3. Test against a differently-oriented stain (not 0°) to check whether n45's advantage generalizes or is specific to this stain geometry.
4. Consider pulling the raw (un-normalized) modality-importance scores (visible as 'score=' text in the dashboard) to confirm whether rising FORCE share reflects force becoming more informative in absolute terms, or IMAGE degrading -- this changes the interpretation of why larger datasets shift the policy's behavior.